

# Finding and Verifying Inverse Functions

## Answer Key

Algebra 2   Composition, Inverses and Transformations

### Quick Answers

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- |                                   |                           |
|-----------------------------------|---------------------------|
| 1. $x - 7$                        | 7. $x^2 + 2$              |
| 2. $\frac{x}{5}$                  | 8. $x, x$                 |
| 3. $\frac{x}{3} + \frac{8}{3}, x$ | 9. 7                      |
| 4. $x, x$                         | 10. $\frac{3x+1}{x-2}, x$ |
| 5. 3                              | 11. -3, -3                |
| 6. $5x + 4$                       | 12. 3                     |
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### Common wrong answers

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5. If they got 43: evaluated  $f$  at 11 instead of the inverse, giving 43  
11. If they got -6.33: divided before subtracting, giving about -6.33 instead of -3
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#### 1. Inverse of $f(x) = x + 7$ .

**Step 1.** Write  $y = x + 7$ , then swap the letters:  $x = y + 7$ . The swap is what turns “output from input” into “input from output”.

**Step 2.** Solve for  $y$ : subtract 7 from both sides, giving  $y = x - 7$ .

**Step 3.**  $f^{-1}(x) = \boxed{x - 7}$

**Check.**  $f(f^{-1}(x)) = (x - 7) + 7 = x \checkmark$

#### 2. Inverse of $f(x) = 5x$ .

**Step 1.**  $y = 5x$  becomes  $x = 5y$  after the swap.

**Step 2.** Divide both sides by 5:  $y = \frac{x}{5}$ .

**Step 3.**  $f^{-1}(x) = \boxed{\frac{x}{5}}$

**Check.**  $f(f^{-1}(x)) = 5 \cdot \frac{x}{5} = x \checkmark$

*Listen for:*  $f^{-1}(x) = \frac{1}{5x}$  — that is the reciprocal of  $f$ , not the inverse function.

#### 3. Inverse of $f(x) = 3x - 8$ .

**Step 1.**  $y = 3x - 8$ ; swap to get  $x = 3y - 8$ .

**Step 2.** Undo in reverse order: add 8 to get  $x + 8 = 3y$ , then divide by 3.

**Step 3.**  $f^{-1}(x) = \boxed{\frac{x + 8}{3}}$

**Check.**  $f\left(\frac{x + 8}{3}\right) = 3 \cdot \frac{x + 8}{3} - 8 = (x + 8) - 8 = \boxed{x}$

*Common wrong answer:*  $\frac{x}{3} + 8$  — the 8 was moved after the division instead of before it.

**4. Verifying  $f(x) = 2x + 9$  and  $g(x) = \frac{x-9}{2}$ .**

**Step 1.** A pair is inverse only if *both* compositions return the input untouched, so compute each one.

**Step 2.**  $f(g(x)) = 2 \cdot \frac{x-9}{2} + 9 = (x-9) + 9 = \boxed{x}$

**Step 3.**  $g(f(x)) = \frac{(2x+9)-9}{2} = \frac{2x}{2} = \boxed{x}$

Both directions give  $x$ , so  $f$  and  $g$  are inverses.

*Why both:* one-sided composition can equal  $x$  on a restricted set while the pair still fails elsewhere. Checking both is the definition, not extra caution.

**5.  $f^{-1}(11)$  for  $f(x) = 4x - 1$ .**

**Step 1.** Find the formula first:  $x = 4y - 1$  gives  $y = \frac{x+1}{4}$ , so  $f^{-1}(x) = \frac{x+1}{4}$ .

**Step 2.** Evaluate at 11:  $\frac{11+1}{4} = \frac{12}{4}$ .

**Step 3.**  $f^{-1}(11) = \boxed{3}$

**Check.**  $f(3) = 4(3) - 1 = 11 \checkmark$

*Common wrong answer:* 43 —  $f$  was evaluated at 11 instead of  $f^{-1}$ .

**6. Inverse of  $f(x) = \frac{x-4}{5}$ .**

**Step 1.** Swap:  $x = \frac{y-4}{5}$ .

**Step 2.** Multiply both sides by 5:  $5x = y - 4$ , then add 4.

**Step 3.**  $f^{-1}(x) = \boxed{5x+4}$

**Check.**  $f^{-1}(f(x)) = 5 \cdot \frac{x-4}{5} + 4 = (x-4) + 4 = x \checkmark$

**7. Inverse of  $f(x) = \sqrt{x-2}$ , domain  $x \geq 2$ .**

**Step 1.** Swap:  $x = \sqrt{y-2}$ .

**Step 2.** Square both sides:  $x^2 = y - 2$ , so  $y = x^2 + 2$ .

**Step 3.**  $f^{-1}(x) = \boxed{x^2+2}$

**Step 4 (the domain).** A square root never returns a negative number, so the outputs of  $f$  — and therefore the inputs of  $f^{-1}$  — are exactly  $x \geq 0$ . The rule  $x^2 + 2$  makes sense for every real number, but as the inverse of this  $f$  its domain is  $x \geq 0$ . Domain and range swap places.

*Listen for:* an inverse quoted as “all reals”. The formula and the function are not the same object; the inverse inherits its domain from the range of  $f$ .

**8. Testing Sofia’s claim:  $f(x) = \frac{x+6}{2}$ ,  $g(x) = 2x - 6$ .**

**Step 1.** Compose in one order:  $f(g(x)) = \frac{(2x-6)+6}{2} = \frac{2x}{2} = \boxed{x}$

**Step 2.** Compose in the other order:  $g(f(x)) = 2 \cdot \frac{x+6}{2} - 6 = (x+6) - 6 = \boxed{x}$

**Step 3.** Both simplify to  $x$ , so Sofia is right: the two functions are inverses.

*Note:*  $g$  undoes “add 6, then halve” by “double, then subtract 6” — the operations reverse and so does their order.

**9.**  $f(x) = x^2$  on  $x \geq 0$ , and  $f^{-1}(49)$ .

**Step 1.** With the domain cut to  $x \geq 0$ , each output comes from exactly one input, so an inverse exists. Swapping gives  $x = y^2$  with  $y \geq 0$ .

**Step 2.** Taking the non-negative root,  $f^{-1}(x) = \sqrt{x}$ , and  $\sqrt{49} = 7$ .

**Step 3.**  $f^{-1}(49) = \boxed{7}$

**Why the restriction matters.** Without it both 7 and  $-7$  square to 49, so “the” input is not defined and  $f$  has no inverse function at all.

**10.** Inverse of  $f(x) = \frac{2x+1}{x-3}$ .

**Step 1.** Swap:  $x = \frac{2y+1}{y-3}$ , then clear the fraction:  $x(y-3) = 2y+1$ .

**Step 2.** Expand and gather every  $y$  term on one side:  $xy-3x = 2y+1$ , so  $xy-2y = 3x+1$ .

**Step 3.** Factor out  $y$  — this is the step the whole problem turns on:  $y(x-2) = 3x+1$ , so  $y = \frac{3x+1}{x-2}$ .

**Step 4.**  $f^{-1}(x) = \boxed{\frac{3x+1}{x-2}}$

**Check.** Substituting into  $f$ :  $\frac{2\left(\frac{3x+1}{x-2}\right) + 1}{\left(\frac{3x+1}{x-2}\right) - 3} = \frac{\frac{6x+2+x-2}{x-2}}{\frac{3x+1-3x+6}{x-2}} = \frac{7x}{7} = \boxed{x} \checkmark$

*Note:* the vertical asymptote of  $f$  is  $x = 3$  and that of  $f^{-1}$  is  $x = 2$  — the excluded input of one is the excluded output of the other.

**11.**  $f^{-1}(-4)$  for  $f(x) = 3x+5$ , two ways.

**Step 1 (formula).**  $x = 3y+5$  gives  $y = \frac{x-5}{3}$ , so  $f^{-1}(x) = \frac{x-5}{3}$  and  $f^{-1}(-4) = \frac{-4-5}{3} = \frac{-9}{3} = \boxed{-3}$

**Step 2 (direct).** Solve  $f(x) = -4$ :  $3x+5 = -4$ , so  $3x = -9$  and  $x = \boxed{-3}$

**Step 3.** The two agree, which is the whole point: asking “ $f^{-1}$  of  $-4$ ” and asking “which input gives  $-4$ ” are the same question.

*Common wrong answer:* about  $-6.33$  — dividing by 3 before subtracting 5. Undo the operations in reverse order.

**12.**  $f(x) = x^2 - 4$ . Part (a) is an open explanation, flagged for manual review; the model wording is below.

**Step 1 (part a).**  $f$  is not one-to-one on the reals:  $f(3) = 5$  and  $f(-3) = 5$ , so the output 5 has two inputs and there is no rule that sends 5 back to a single number. Graphically, a horizontal line meets the parabola twice, so the reflection across  $y = x$  is not a function.

**Step 2 (part b).** Restricting to  $x \geq 0$  keeps only the right half of the parabola, which passes the horizontal-line test. Swapping gives  $x = y^2 - 4$ , so  $y^2 = x + 4$  and, taking the non-negative root,  $f^{-1}(x) = \sqrt{x+4}$  on  $x \geq -4$ .

**Step 3.**  $f^{-1}(5) = \sqrt{5+4} = \sqrt{9} = \boxed{3}$

**Check.**  $f(3) = 9 - 4 = 5 \checkmark$

*Full credit:* part (a) must say that two different inputs share an output (or invoke the horizontal-line test), not merely that “the graph is a parabola”. Part (b) needs the restriction stated, the formula  $\sqrt{x+4}$ , and the value 3.